

Nguyen Thien Loc

AI Engineer

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SUMMARY

Final-year Computer Science student at HCMUT with hands-on experience in **LLM Systems & Multi-Agent Architectures** (LangGraph, MCP, A2A, RAG) and **Computer Vision / Deep Learning** (Adversarial Training, Generative Models). Backed by enterprise training at VinUniversity and production internship experience, seeking to leverage strong modeling foundations, software engineering, and an AI product mindset to build scalable, high-impact intelligent systems in production.

EXPERIENCE

VINUNIVERSITY - AI in Action Program Cohort 3

Jul. 2026 – Present

AI Talent (Track 2: AI Infrastructure)

Ho Chi Minh City, Vietnam

- Completed an intensive 12-week program by **Vingroup & VinUniversity**, built around global **SFIA standards** and an end-to-end AI product mindset.
- Phase 1 (Foundations & Product)**: Built and evaluated LLM application pipelines, **Multi-Agent workflows (MCP, A2A)**, Prompt Engineering, and AI Safety/Guardrails.
- Phase 2 (Track 2: AI Infrastructure)**: Provisioned cloud AI infrastructure on **GCP** via **Terraform**, engineered **Data Lakehouses** (Delta Lake, Iceberg), and managed **Vector/Feature Stores** (Qdrant, Feast).

bTasker

May 2026 – Jul. 2026

AI Engineer Intern

Ho Chi Minh City, Vietnam

- Developed a **Hybrid RAG pipeline** (Dense + BM25 Sparse via **Qdrant & LlamaIndex**) to accurately match unstructured customer complaints with relevant Tasker training modules.
- Built a **FastMCP tool server** that standardized issue extraction, vector retrieval, and LLM verification for autonomous multi-agent execution.
- Orchestrated **Agent-to-Agent (A2A)** multi-agent workflows on an asynchronous Kanban engine to automate test-case generation and Playwright execution.
- Engineered robust **LLM pipelines** with strict JSON schema validation, provider fallback mechanisms, and **Langfuse** observability.

DFM Engineering

Apr. 2025 – Jul. 2025

AI Engineer Intern

Ho Chi Minh City, Vietnam

- Fine-tuned a **Pix2Pix (cGAN)** model to translate 2D architectural layouts into scaffold structure maps, supporting automated construction planning.
- Developed **C# AutoCAD automation scripts** to generate **3D slicing workflows**, reducing manual modeling effort.

Grab

Apr. 2025 – May 2025

Grab Tech Bootcamp 2025

Ho Chi Minh City, Vietnam

- Unipath**: Developed an exam analytics platform for score prediction and personalized university admission recommendations.
- Built an end-to-end ETL pipeline using **Apache Airflow, Selenium, and BeautifulSoup** to automate large-scale data collection and processing.
- Trained and optimized an **XGBoost** regression model for admission cutoff prediction, achieving **Top 4/14 teams**.

PROJECTS

ViVi: Multi-Agent Test-Drive Booking Assistant | [🔗 Live Demo](#) | *FastAPI, LangGraph, PostgreSQL, React, GCP*

- Architected a stateful **LangGraph multi-agent system** with skill-based routing, intent classification, and sensitive policies coordinating a 9-step booking state machine.
- Implemented real-time concurrency with **atomic slot reservations** (Redis/PostgreSQL), SSE token streaming, and PostgreSQL checkpoints for persistent conversation memory.
- Engineered a **4-layer AI Guardrail** (prompt injection defense, anti-hallucination, LLM budget, 3-strike ban) and deployed multi-cloud infrastructure on **GCP & Cloudflare**.

Capstone Project: Adversarial Training for Face Super-Resolution | *Python, PyTorch, CUDA*

- Conducted a comprehensive benchmark of state-of-the-art **Face Super-Resolution** models and categorized existing approaches based on architectural design and learning strategies.
- Proposed an **adversarial training strategy** to balance perceptual realism and identity preservation, integrating **SPARNet** and **CTCNet** in a plug-and-play manner.
- Evaluated on **CelebA, HELEN, and FFHQ** using **PSNR, SSIM, LPIPS, and FID**, demonstrating improved perceptual quality and identity preservation across ablation studies.

EDUCATION

Ho Chi Minh City University of Technology (VNU - HCMUT)
Bachelor of Computer Science

Sep. 2022 – Oct. 2026
Ho Chi Minh City, Vietnam

- Outstanding Academic Performance 2024 – 2025

ACTIVITIES

BKFC - Bui Thi Xuan Admissions Advisory |  Fanpage

2022 – 2023

Team Leader

Ho Chi Minh City, Vietnam

- Led a team of **15+ members** managing social media channels, creating orientation content, and hosting engagement events for high school students.
- Coordinated university admissions counseling sessions at high schools and organized on-campus tour programs for visiting student delegations.

SKILLS

Languages & Frameworks: Python, C++, SQL, PyTorch, FastAPI, Scikit-learn

LLM & Agent Systems: LangGraph, Model Context Protocol (MCP), Agent-to-Agent (A2A), RAG, LlamaIndex, AI Guardrails

Databases & Observability: PostgreSQL, Redis, Qdrant, Langfuse

Cloud & DevOps: Google Cloud (GCP), Terraform, Docker, Playwright, CI/CD, Linux

Languages: English (IELTS 6.0)